Chengze Li

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RESEARCH EXPERIENCE

Manteia Technologies Co., Ltd. Milwaukee, WI

Research Scientist 04/2025-06/2025

Research Scientist Intern 05/2024-09/2024

- Architected the core geometry engine for the AccuContour system to deliver multi-organ 3D segmentation with submillimeter boundary precision that serves as the strictly constrained input for safety critical radiation dose planning in daily clinical operations, reducing manual contouring workload by 35%.
- Overcame physical reality gaps including breathing motion and metal artifacts by implementing a deformable registration module with anatomical regularization that maintained topology consistency and prevented mesh shearing across 1000 variable quality scans by modeling sliding interfaces between tissues.
- Built explainable, clinically oriented tooling for the segmentation pipeline, including contour-stability constraints on volume and centroid location that demonstrated a 28% reduction in shape drift when benchmarked against AccuContour's validated models using Dice and HD95. Additionally, integrated a retrieval-augmented VLM module that surfaces patient-specific risk factors and supporting oncology evidence for adaptive planning and QA.

University of California, Los Angeles, *Research Assistant* Los Angeles, CA. 09/2024-04/2025

- Engineered a multimodal 3D prediction pipeline processing over 900 patient CT volumes that replaces standard neural networks with quantitative voxel-based feature extractors to fuse clinical metadata with fine-grained structural anomalies by explicitly modeling 3D shape topology.
- Solved critical domain shift failures caused by variable scanner resolutions across 9 independent hospitals by implementing isotropic resampling to a unified 1.0 mm grid to enforce strict physical scale invariance, decoupling anatomical features from sampling artifacts to prevent overfitting to hardware-specific noise.
- Deployed a production ready quality gate using Platt Scaling that recalibrated risk probabilities to achieve a reliable AUC score above 0.85 under population shift, and built an explainability layer that maps abstract features into auditable physical concepts such as tissue heterogeneity for doctor review.

EDUCATION

University of Illinois Chicago – *College of Engineering* 08/2025-09/2029

- *Ph.D. Student in Computer Science* Advisor: Philip S. Yu

Columbia University - *Fu Foundation School of Engineering* 09/2023-03/2025

- *Master of Science, Applied Mathematics* Advisor: Kui Ren

University of North Carolina Wilmington - *College of Arts and Sciences* 09/2019-05/2023

- *Bachelor of Science, Mathematics* (Joint Dual Degree Program with Chongqing University of Arts and Sciences)

PUBLICATIONS

- **Chengze Li**, Xiao Liu, …, Sharon Qi, Philip Yu, "Closing the Domain Gap: Geometric Invariant Learning and Post-Hoc Probability Calibration for Robust 3D Radiomics Survival Prediction". (Submitted to MICCAI 2026)
- **Chengze Li**, Zirong Li, …, Qichao Zhou, Philip Yu, "Physics-Guided Spatiotemporal Learning: Fusing Longitudinal CBCT and 3D Dose Distribution with Spatial Attention for Early Pneumonitis Prediction". (Submitted to MICCAI 2026)
- Tianyang Xu, Haojie Zheng, **Chengze Li**, …, Lichao Sun "NodeRAG: Structuring Graph-based RAG with Heterogeneous Nodes". https://arxiv.org/pdf/2504.11544.
- Zhenjiang Li, Wei Zhang, Baosheng Li, Jian Zhu, Yinglin Peng, **Chengze Li**, Jennifer Zhu, Qichao Zhou, Yong Yin, "Patient-specific Daily Updated Deep Learning Auto-Segmentation for MRI-Guided Adaptive Radiotherapy", Radiotherapy and Oncology, Dec. 2022, Vol.177, P222-230